

From High-Dimensional Weather Data to Accurate Renewable Generation Forecasts: A Statistical Aggregation Method

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Abstract—Accurate short-term photovoltaic (PV) and wind power forecasting relies heavily on weather information across different locations. In practice, however, directly incorporating high-resolution numerical weather prediction (NWP) data from numerous grid points results in high-dimensional feature sets, which may increase computational burden and deteriorate the generalization capability of forecasting models. Against this background, we introduce a statistical spatial aggregation approach that summarizes each weather variable, such as wind speed, and irradiance, into six descriptive statistics, namely mean, standard deviation, minimum, maximum, skewness, and kurtosis, computed across all spatial grid points. This compact representation preserves essential spatial characteristics while drastically reducing the dimensionality of the input space. The efficacy of the proposed method is demonstrated using real-world PV and offshore wind datasets in conjunction with eight different weather forecast providers. Comparative results indicate consistent improvements in both deterministic and probabilistic forecasting performance over common baselines, including single-point inputs and simple spatial averaging. Finally, the proposed approach is model-agnostic, scalable, and readily integrable into operational forecasting pipelines.

Index Terms—renewable energy forecasting, spatial weather aggregation, solar power, wind power, weather forecasts

I. INTRODUCTION

The increasing penetration of renewable energy sources, such as photovoltaic (PV) and wind power, presents new challenges for electricity markets and grid operators, who require accurate and reliable forecasts of power generation to ensure grid stability and efficient market operations. Traditional PV and wind power forecasting methods often utilize weather data from a single location/weather station or apply simple averaging across multiple sites. Those approaches may fail to capture the spatial diversity and variability of weather conditions influencing geographically dispersed renewable assets [1]. Furthermore, directly incorporating high spatial resolution weather data from numerous grid points leads to high-dimensional input spaces, which can increase computational costs and may reduce the efficiency and generalizability of forecasting models [2].

Recent studies highlight the value of leveraging spatial-temporal dependencies among geographically distributed renewable generation sites to improve short-term forecasting

performance. Most this work is grounded in vector autoregressive formulations, in which the power output at each site is modeled using its own historical measurements together with lagged observations from neighboring locations [3]–[5]. Although these developments represent meaningful progress, the underlying approaches predominantly exploit spatial correlations captured directly in measured power time series, which might not be always available in real-world applications due to different PV/wind park ownership or other data unavailability reasons. Furthermore, those methodologies have been tested only in very short term horizons, e.g., 6 hours ahead, and not for day-ahead, e.g., 36 hours ahead.

Another type of approach has been proposed in [2], where the authors introduced an algorithm to select the most relevant weather stations to get the weather forecasts from. Similar approaches were proposed from the winning teams in the Global Energy Forecasting Competition 2014 (GEFCom2014) [6], where weather data from different zones, i.e., PV and wind parks, were combined to forecast the power generation of a specific zone [7]–[10]. Nevertheless, those approaches are restricted by the number of weather stations that can provide weather forecasts and does not consider a grid of numerical weather predictions (NWP).

To this end, this work introduces a method for spatial aggregation of a grid of NWP, specifically designed to enhance short-term forecasting of PV and wind power generation. The proposed approach employs statistical feature extraction to summarize key weather variables, e.g., wind speed, solar radiation, temperature, etc., across multiple grid points using a set of statistical descriptors. This compact representation preserves both the magnitude and spatial variability of weather conditions, enabling forecasting models to retain critical information while significantly reducing input dimensionality and thus, computational costs.

II. PROPOSED METHOD

The objective of the proposed methodology is to compress high resolution, multi-site NWP data into a compact feature representation that retains the spatial characteristics that are relevant for PV and wind power forecasting. Rather than supplying forecasting models with raw values from numerous grid points, which substantially increases dimensionality and

often impairs model generalization, we aggregate each meteorological variable across all available spatial locations for any given forecast time step t . This yields a concise set of spatial summary descriptors that capture both the overall magnitude and the inherent heterogeneity of the underlying weather field. In doing so, the resulting representation preserves essential spatial structure while ensuring that downstream forecasting models remain tractable, robust, and less susceptible to overfitting.

For each meteorological variable x observed across n spatial grid points at a given forecast time step, we replace the raw multi-site values with a compact set of six statistical descriptors. Let $x_1, \dots, x_n$ denote the values of a variable, e.g., wind speed, irradiance, temperature, humidity, pressure, across the grid, and let $\bar{x}$ and σ be the sample mean and sample standard deviation, respectively. We compute: the mean to capture the central tendency; the standard deviation to quantify spatial variability and gradients; the minimum and maximum to retain localized extremes; and the third and fourth standardized moments, i.e., skewness and kurtosis, to encode asymmetry and tail heaviness of the spatial distribution. This yields a fixed-length representation of size $6 \times V$, where V is the number of meteorological variables, providing a distribution-aware yet low-dimensional summary which is suitable for downstream deterministic or probabilistic forecasting models.

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

$$\sigma = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} \quad (2)$$

$$x_{\min} = \min\{x_1, x_2, \dots, x_n\} \quad (3)$$

$$x_{\max} = \max\{x_1, x_2, \dots, x_n\} \quad (4)$$

$$\text{Skew}(x) = \frac{1}{(n-1)\sigma^3} \sum_{i=1}^n (x_i - \bar{x})^3 \quad (5)$$

$$\text{Kurt}(x) = \frac{1}{(n-1)\sigma^4} \sum_{i=1}^n (x_i - \bar{x})^4 \quad (6)$$

This descriptor set strikes a deliberate balance between simplicity and expressive power. The mean characterizes the large-scale background state of the meteorological field, e.g., prevailing regional wind speeds or irradiance levels, while the standard deviation reflects spatial gradients and variability that are often responsible for aggregate ramping events and forecast uncertainty. The minimum and maximum retain information about localized conditions, such as calm wind pockets or high-irradiance zones, that can have an outsized influence on system-level production, particularly in cases where the installed capacity is spatially dispersed. Higher-order moments,

including skewness and kurtosis, encode asymmetries and tail behavior, capturing phenomena such as sharp cloud boundaries or wind gusts that are not adequately represented by measures of central tendency alone. Hence, these statistics provide a compact yet distribution-aware summary of the spatial field, delivering richer informational content than simple averaging while avoiding the fragility associated with selecting individual grid points.

The aggregation reduces the input dimensionality from $n \times V$ raw grid-point values, which might often reach thousands of inputs in case of system-level forecasting (e.g., 200 grid points $\times$ 10 weather variables), to a compact set of $6 \times V$ descriptors, effectively shrinking the feature space by one to two orders of magnitude. This compression has several implications. First, it enhances learning efficiency and mitigates overfitting risks for statistical and machine learning models by removing redundant spatial detail while retaining the dominant structure of the NWP field. Second, it leads to shorter training and inference times and a substantially lower memory requirements. Those technical characteristics might be essential in operational environments where multiple forecast providers and frequent update cycles must be processed quickly and reliably. Third, the approach improves model portability where the same compact interface supports different NWP vendors or evolving grid configurations without the need to retrain feature extractors tied to specific geographical layouts. Finally, this method is model-agnostic and integrates seamlessly into existing forecasting pipelines.

III. DATA & EXPERIMENTAL SETUP

The proposed spatial aggregation framework is assessed in the context of short-term forecasting for system-level PV and wind power generation. In both cases, the forecasting target



Fig. 1. Offshore wind: Map of the study region with the NWP grid [11].

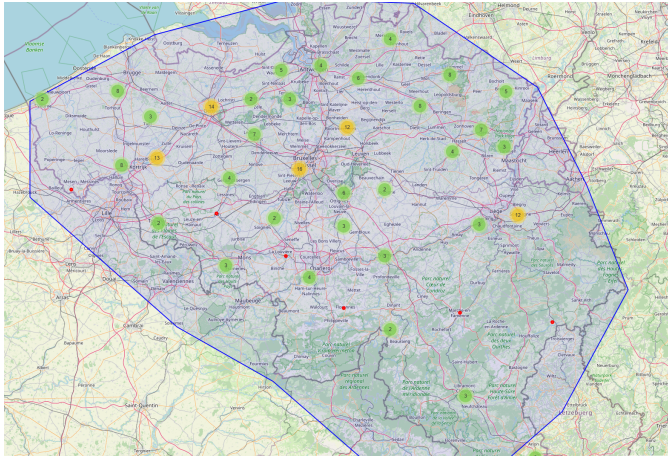


Fig. 2. PV: Map of the study region with the NWP grid [11].

corresponds to the aggregated output of multiple geographically dispersed assets within a defined region, reflecting the system-level perspective relevant for operational planning and short-term markets. Starting with wind power, the evaluation focuses on day-ahead forecasting with a 15-minute resolution, aligning with the temporal granularity typically required by system operators and market participants. An analogous day-ahead horizon is adopted for the PV experiments, ensuring a consistent temporal structure across technologies and enabling a fair assessment of the method's performance under different renewable generation technologies. This setup allows us to isolate the effect of the spatial descriptor representation while maintaining comparability across the two resource types.

A. Wind power forecasting

The wind power experiments are based on real-world data from the Belgian offshore wind fleet, providing a setup that closely reflects the operational conditions of a transmission system operator (TSO). System-level generation measurements are sourced from ELIA's publicly available Open Data platform [12], which offers historical aggregated offshore wind production time series together with their planned or forced availability due to maintenance or other unexpected events. These observations are paired with NWP from different weather forecast providers that are described below. Each provider supplies spatially resolved forecasts for key meteorological variables, including wind speed, wind direction, temperature, and pressure, over a grid covering the relevant offshore region. To form the grid, a polygon was designed around each offshore wind park and then, a grid of $2\text{ km} \times 2\text{ km}$ was created within each polygon. Fig. 1 illustrates the NWP grid over the offshore wind farms of Belgium based on OpenStreetMap [11]. By combining multi-provider NWP inputs with aggregated system-level production data, the experiments enable a comprehensive assessment of how the proposed spatial aggregation approach influences forecasting accuracy and robustness under diverse weather information sources.

B. PV power forecasting

To further assess the generality of the proposed spatial aggregation approach, a parallel experimental setup is implemented for system-level PV power forecasting. As in the wind power forecasting case, spatially resolved NWP data serve as predictors, while the target variable corresponds to aggregated PV generation of Belgium obtained from [12]. To generate the NWP grid, a different approach is employed, since PV capacity is not distributed uniformly across Belgium. Instead, we overlay a fine spatial grid on the study area, aggregate the installed PV capacity within each grid cell, and convert these capacities into sampling probabilities proportional to each cell's share of total installed PV capacity. A set of 200 grid cells is then drawn according to these probabilities, and their centroids are used as the effective NWP points. Fig. 2 depicts the derived NWP grid points. For visibility reasons, several grid points are visually aggregated and represented by green and yellow circles that indicate the number of points included in those small regions.

Evaluating the total PV power generation of a country allows us to showcase the applicability of the proposed method to a large and geographically diverse region. The PV experiments utilize radiation-related variables, such as shortwave, direct, and diffuse radiation, which play a dominant role in short-term PV variability. Given the strong sensitivity of PV output to cloud-driven spatial dynamics, this setting provides an additional and complementary test bed for evaluating the ability of the compact descriptor framework to capture spatial heterogeneity in meteorological conditions. This parallel design enables a consistent comparison across renewable technologies while validating the method under fundamentally different weather regimes.

C. Weather forecast providers

Across both the PV and wind power case studies, the same set of eight weather forecast providers is employed to ensure a consistent and comparable evaluation framework. While the providers vary in spatial resolution, physical modeling approaches, and update frequencies, each one supplies a list of relevant meteorological variables required for constructing the proposed statistical descriptors. For every weather forecast provider, raw values are extracted over the respective area without any specific tuning or spatial weighting. This design

TABLE I
WEATHER MODELS EMPLOYED IN THIS WORK.

Weather Model	National Weather Provider	Origin Country	Spatial Res. (km)	Forecast Length (days)
ICON	DWD	Germany	2	7.5
GFS	NOAA	USA	13	16
AROME	Météo-France	France	2.5	4
IFS	ECMWF	EU	25	15
UKMO	UK Met Office	UK	2	7
MET Nordic	MET Norway	Norway	1	2.5
HARMONIE	KNMI	Netherlands	2	2.5
HARMONIE	DMI	Denmark	2	2.5

ensures that the comparative analysis isolates the impact of the spatial aggregation method itself rather than artifacts introduced by preprocessing choices. The resulting data are subsequently transformed into the feature vectors described previously, yielding a unified and low-dimensional input representation suitable for the downstream forecasting models. Table I lists the eight weather forecast providers employed in this work together with their corresponding weather model, origin country, spatial resolution, and forecast length [13].

D. Forecasting model

All forecasting experiments employ the same LightGBM model in order to isolate the effect of the input representation [14]. The same hyperparameter configuration and training parameters are used across all weather forecast providers, ensuring strict methodological consistency. It also worth pointing out that apart from the deterministic PV and wind power forecasting models, two additional probabilistic models are trained to predict the 10th and 90th quantile for both PV and wind power. This is done by setting the objective function of LightGBM to the respective quantile error.

Apart from the spatial aggregation, a temporal preprocessing is also required as the power generated at any time step is highly correlated with the power generated during the previous time steps. To capture those autocorrelation characteristics, a rolling average for a window of three and five hours is calculated for the wind speed related variables (mean, std., etc.) for wind power forecasting and for the solar radiation variables for PV power forecasting, respectively. In addition, the previous and next value is extracted for each weather variable.

E. Evaluation

Historical data from 1st January 2023 to 31st December 2024 are used for training while the test set comprises data from 1st January 2025 to 31st May 2025. Importantly, discrete day-ahead predictions are sequentially generated for each date. The deterministic accuracy is quantified using the root mean squared error (RMSE) as:

$$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{t=1}^N (y_t - \hat{y}_t)^2}, \quad (7)$$

where T denotes the total number of time steps forecasted, y_t is the real measured power, and $\hat{y}_t$ the corresponding forecast. The probabilistic performance is evaluated through the Mean Winkler Interval (MWI). For each time step t , the Winkler Interval is calculated as:

$$\text{WI}_t = \begin{cases} (U_t - L_t) + \frac{2}{\alpha}(L_t - y_t), & \text{if } y_t < L_t \\ (U_t - L_t) & \text{if } L_t \leq y_t \leq U_t \\ (U_t - L_t) + \frac{2}{\alpha}(y_t - U_t), & \text{if } y_t > U_t \end{cases} \quad (8)$$

where U and L represent the 90th and 10th quantile, respectively, and α is the nominal miscoverage probability of the prediction interval, i.e., 0.2 in this case. The WI evaluates the quality of prediction intervals by rewarding narrow intervals

that successfully contain the true value while imposing increasing penalties when observations fall outside the predicted bounds.

IV. RESULTS

To thoroughly evaluate the performance and the robustness of the proposed method, we compare both its deterministic and probabilistic performance with two commonly used alternatives from the literature: 1) using weather forecasts from a

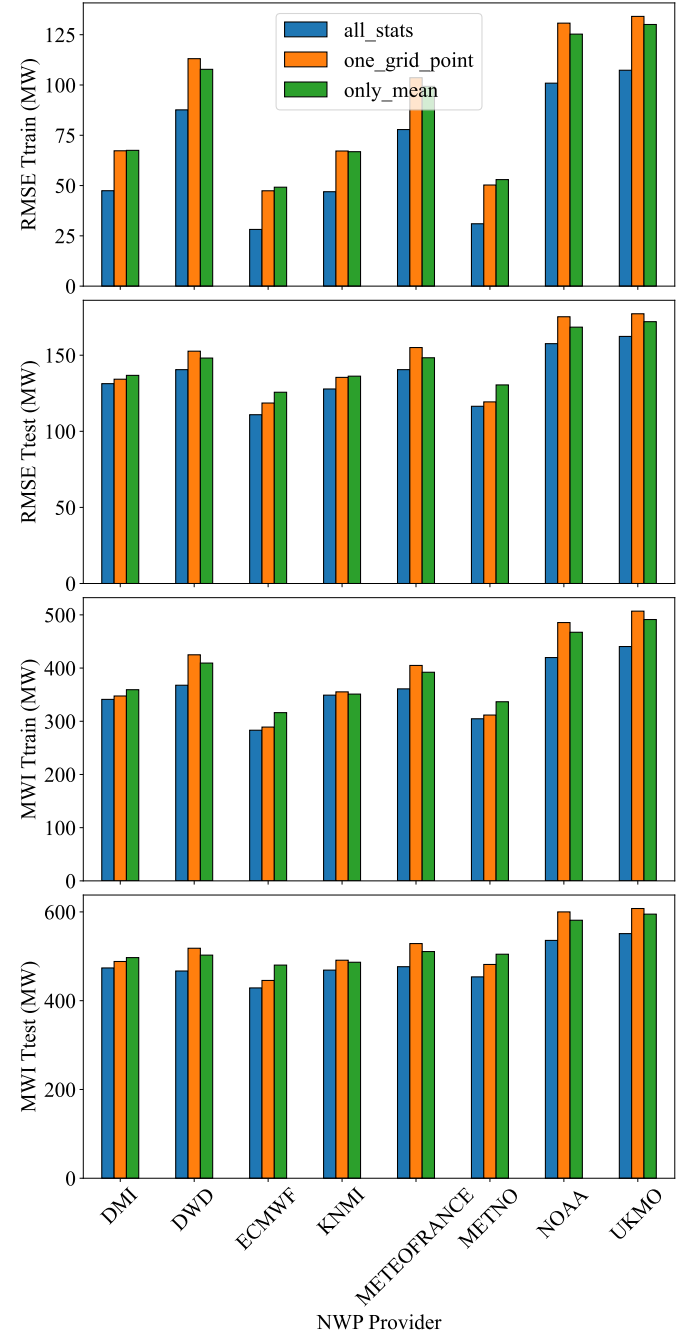


Fig. 3. Offshore wind: Deterministic (RMSE) and probabilistic (MWI) metrics for different NWP providers and weather data processing techniques.

single grid point ("*one_grid_point*") [15], and 2) using only the mean of all grid points ("*only_mean*") [8].

Starting with wind power forecasting, Fig. 3 illustrates the average RMSE and MWI for both training and test sets over the three different weather data processing techniques. Note that "*all_stats*" correspond to the proposed aggregation method, "*one_grid_point*" considers NWP only from the centroid of the region under study, and "*only_mean*" estimates only the average value across the NWP grid points for each weather variable.

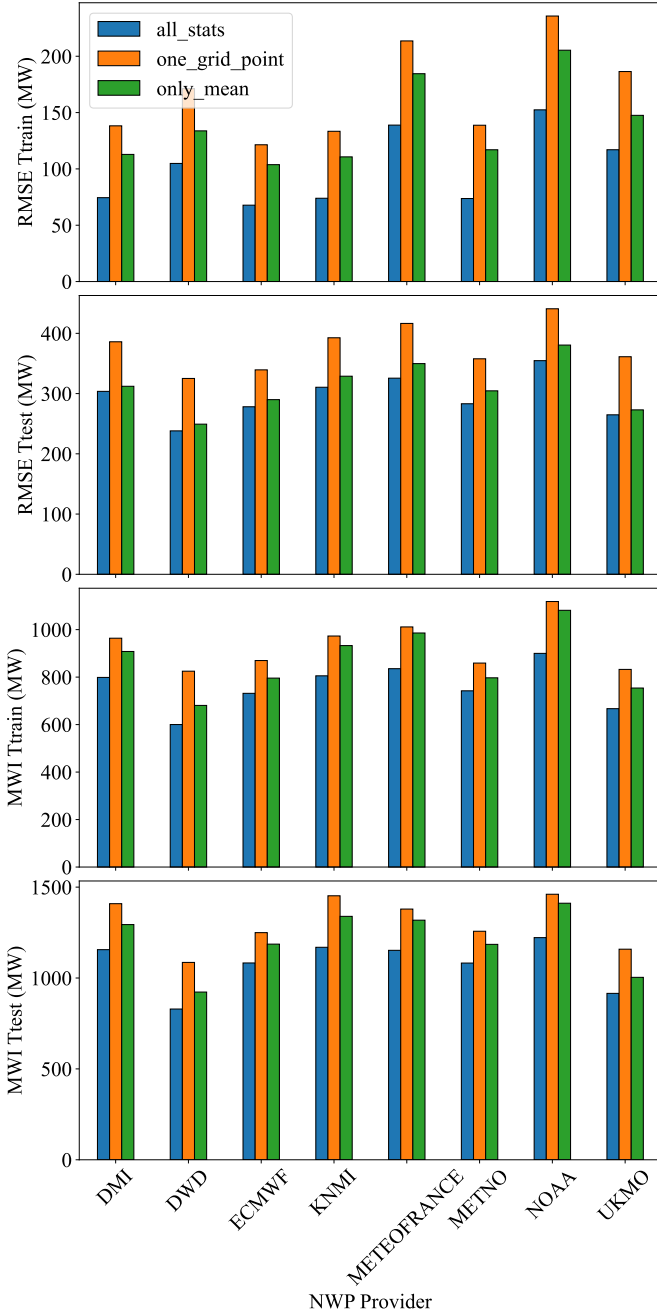


Fig. 4. PV: Deterministic (RMSE) and probabilistic (MWI) metrics for different NWP providers and weather data processing techniques.

The proposed method achieves the lowest RMSE on both the training and test datasets for every provider, as shown in Fig. 3. These reductions in deterministic error underscore the value of representing not only average conditions but also the spatial variability and distributional structure of the wind-related features that the competing methods could not capture. Similar improvements are observed in the probabilistic evaluation. The Winkler scores reported Fig. 3 indicate that the prediction intervals produced by the model trained based on the proposed method are systematically sharper and better calibrated, demonstrating enhanced uncertainty quantification and more informative probabilistic forecasts.

A nearly identical performance pattern is observed in the PV forecasting experiments. The RMSE values for both the training and test sets (Fig. 4) again indicate that the full statistical descriptor approach consistently outperforms the other two baselines. Furthermore, the Winkler scores reported in Fig. 4 show that the model trained based on the proposed method produces tighter and more reliable prediction intervals, demonstrating a better probabilistic performance across diverse sky conditions.

V. CONCLUSION

The experimental results presented in this work show that summarizing spatially distributed NWP inputs through statistical descriptors yields a more informative representation for system-level renewable power forecasting compared to the commonly used baselines. The consistent performance gains across all eight weather forecast providers indicate that the proposed approach is not dependent on any particular NWP provider, physical model, or regional configuration. Instead, it captures fundamental spatial characteristics, such as gradients, localized extremes, and higher order distributional asymmetries, that simple averaging or single point values cannot encode. Equally important, the proposed representation reduces the input dimensionality by orders of magnitude and stabilizes training even under noisy or heterogeneous meteorological conditions. Nonetheless, the proposed method's advantage may diminish in scenarios with limited spatial variability, or small regions in general.

It is worth pointing out that this work has been already deployed in production since several months and daily delivers accurate and sharp predictions. Future work will focus on more experiments including ablation studies to understand the influence of each statistical descriptor on the prediction improvements. In addition, the sensitivity of this work to different number of grid points will be studied in an attempt to find a "threshold" after which, the proposed method outperformed the commonly used baselines.

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